Solutions to Exercises after Class 5.

2. Let X_1 and X_2 be the waiting times of the two people being served. Then $X_1, X_2 \sim E(\lambda)$ and X_1, X_2 are independent. We have to wait $Y = \min(X_1, X_2)$ time which is again exponentially distributed with parameter 2λ (see material from Class 5). Therefore we have to wait $\frac{1}{2\lambda} + \frac{1}{\lambda} = \frac{3}{2} \cdot \frac{1}{\lambda}$ time.

1. Let A_1 be the event that the first point is in one of the three intervals, A_2 be that the first two points fall into different intervals and A_3 that all the 3 points fall into different intervals. Then

$$\mathbb{P}(A_3) = \mathbb{P}(A_1 A_2 A_3) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2 | A_1) \mathbb{P}(A_3 | A_1 A_2) =$$

$$= 1 \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}.$$